MILITARY SPECIFICATION

WHEEL, ABRASIVE, ALUMINUM OXIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mandrel mounted abrasive wheels suitable for trimming synthetic resin denture base materials.

1.2 Classification. The abrasive wheels shall be of the following shapes as specified (see 6.2).

Shape 1 - Cone
Shape 2 - Flame
Shape 3 - Bud

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation-packaging, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-66 - Steel, Chemical Composition and Hardenability.
FED-TEST-METHOD STD-No. 151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B74.12 - Specification for the Size of Abrasive Grain-Grinding Wheels, Polishing and General Industrial Uses.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive. The abrasive shall be 54 grit, high quality, white, aluminum oxide produced in the arc type electric furnace. The grain size shall conform to American National Standards Institute specification B74.12 for abrasive grain size.

3.1.2 Binder. The binder shall be high quality porcelain type vitrified binder. The abrasive and binder shall be of proper proportions and mix and be properly fired to form grinding wheels suitable for the purpose intended.

3.1.3 Mandrels. The mandrels shall be fabricated from corrosion resisting steel conforming to AISI 416 of FED-STD-66, when tested as specified in 4.5.5.

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3.1.4 Adhesive. The grinding wheels shall be securely mounted to the mandrels with high quality epoxy or phenolic adhesive.

3.2 Style, design and dimensions. The mounted wheels shall be free cutting, self-cleaning and suitable for trimming synthetic resin denture base materials with a minimum of heat generation. The wheels shall be of medium hardness and of a dense structure (see 6.4). The wheels shall meet the requirements specified herein and shall conform to the dimensions and shapes specified in figure 1.

3.3 Runout. The runout shall not exceed 0.003 inch overall, when tested as specified in 4.5.2.

3.4 Boil test. The mandrels shall show no signs of corrosion when subjected to the boil test specified in 4.5.3.

3.5 Performance. The abrasive wheels shall trim resin denture base material evenly with no deep scratches or coarse rough spots when tested as specified in 4.5.4. The abrasive wheels shall not generate excessive heat, shall have self cleaning properties and shall not show excessive disintegration when tested as specified in 4.5.4.

3.6 Finish. The mandrels shall have a bright smooth finish.

3.7 Count. Unless otherwise specified (see 6.2), six wheels of one shape number shall constitute one unit of issue.

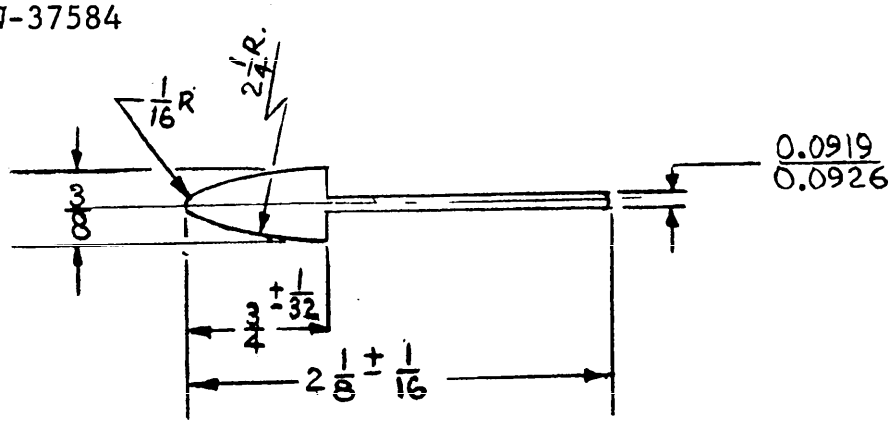
3.8 Workmanship. Workmanship shall be first class throughout. Wheels shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

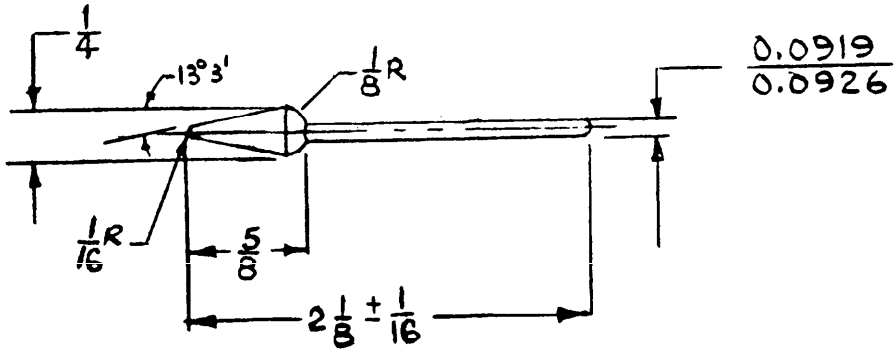
4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

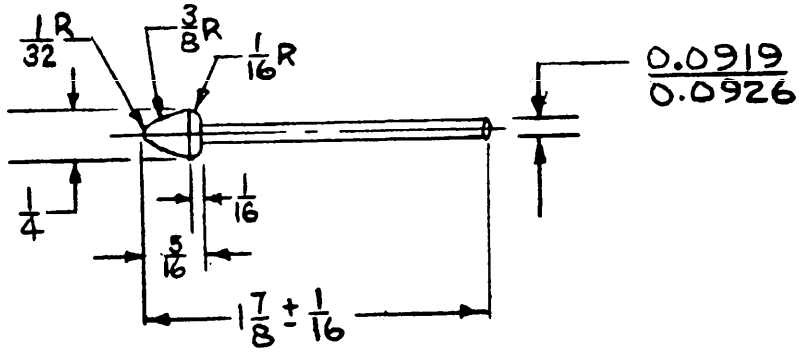
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SHAPE 1-CONE



SHAPE 2-FLAME



SHAPE 3-BUD

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCES ARE AS FOLLOWS

FRACTIONS ± 1/64

ANGLES ± 1/4°

FIGURE 1. Wheel, Abrasive.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the abrasive for grain size and steel for composition may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material/component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Certificate of compliance. Certificates of compliance for the type of abrasive, type of binder, hardness and structure of the wheels may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one wheel of each shape (see 1.2).

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.5
Minor	II	4.0
Dimensional	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, table II, 4.2.2.1.1 and 4.2.2.1.2.

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TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Unit of product	Inspection level	AQL (percent defective)
Runout	3.3	4.5.2	1 wheel	S-1	1.5
Boil test	3.4	4.5.3	1 wheel	S-1	1.5
Performance	3.5	4.5.4	1 wheel	S-1	1.5

4.2.2.1 Raw material.

4.2.2.1.1 A sufficient quantity of abrasive material used in the manufacture of the wheels shall be tested for particle size in accordance with 4.5.1.

4.2.2.1.2 A sufficient quantity of each lot of metal used in the manufacture of the mandrels shall be selected for composition analysis in accordance with 4.5.5.

4.3 Examination

4.3.1 Abrasive wheels. The abrasive wheels shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination shall include, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Abrasive grit not as specified.
102	Wheels not the required shape.
103	Mandrel not centered in wheel.
104	Mandrel not securely bonded to wheel.
105	Mandrel not free of sharp edges.
106	Mandrel bent.
Minor	
201	Incorrect count in unit of issue.

4.3.2 Dimensional examination. The wheels shall be examined for defects in dimensions. Any dimension not within the tolerance specified in figure 1 shall be classified as a defect.

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4.4 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests.

4.5.1 Abrasive particle size. The determination of grain size shall be made on a sufficient amount of abrasive from each lot, in accordance with procedures set forth in ANSI B74.12.

4.5.2 Runout. The wheel shall be held in a vee-block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 in. equipped with a knife edge foot, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the wheel and shall be so positioned that the foot contacts the wheel neck at the point of smallest diameter. The difference between the maximum and minimum dial indicator readings for one full rotation of the wheel shall be taken as the total indicated runout.

4.5.3 Boil test. The wheels shall be scrubbed with soap and water, rinsed thoroughly with distilled water and dried. Immediately after cleaning, the wheels shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for an additional 24 hours. The wheels shall be removed from the water, allowed to dry, and examined for corrosion.

4.5.4 Performance test. The abrasive wheels shall be used to grind and trim a piece of acrylic resin denture base material approximately 64 millimeters square with a minimum thickness of 5 millimeters. The grinding cycle shall be 10 sec grinding, 20 second not grinding for a total accumulated time of 2 minutes. Visual inspection shall be made of the wheel for conformance with 3.5.

4.5.5 Composition analysis. Tests for composition analysis shall be conducted in accordance with method 111 or 112 of FED-TEST-METHOD-STD-NO. 151.

5. PACKAGING

5.1 Preservation packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One package containing six wheels of the shape specified constitutes one unit of issue.

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5.1.2 Level A.

5.1.2.1 Unit packaging. Six wheels of one shape only shall be thoroughly clean, dry and packaged in a box conforming to PPP-B-566, PPP-B-676 or in a plastic box having a hinged cover and a friction closure. The box shall contain suitable interior fittings so designed to prevent movement of or damage to the wheels.

5.1.2.2 Intermediate package. Twelve unit packages containing wheels of one shape only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Wheels shall be clean, dry and six wheels of one shape only shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages (288 units) containing wheels of one shape only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of box specification.

5.2.2 Level B. Twenty-four intermediate packages (288 units) containing wheels of one shape only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged wheels shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. In addition, marking shall include the shape number as specified in 1.2.

5.3.2 Exterior or shipping containers. The exterior or shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The abrasive wheels covered by this specification are intended to be used to trim synthetic resin denture bases during dental prosthetic laboratory procedures.

6.2 Ordering data. Procurement documents.

(a) Title, number and date of this specification.
(b) Wheel shape number (see 1.2).
(c) When count to be other than specified in 3.7.
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock Number	Item identification
6520-00-880-6842	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Cone 3/8 by 3/4 inch, 6s
6520-00-880-6843	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Flame, 1/4 by 5/8 inch, 6s
6520-00-880-6844	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Bud, 1/4 by 5/16 inch, 6s

6.4 Wheels covered by this specification are identifiable as A54N1V in accordance with ANSI B74.13. Markings for Identifying Grinding Wheels and Other Banded Abrasives.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6520-1437

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-W-37584 WHEEL, ABRASIVE, ALUMINUM OXIDE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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